

PRESENTED BY :

Tarek Eltantawy

3D SHAPE COMPLETION WITH AN ENCODER- PREDICTOR NETWORK

A lightweight 3D shape completion pipeline for filling missing geometry in partial 3D scans using voxel-based TSDF input and an EPN-style neural model.

OBJECTIVE

To develop a reproducible 3D shape completion pipeline that reconstructs full 3D objects from partial TSDF inputs using an Encoder–Predictor Network (EPN). The goal is to explore the relationship between training data types (synthetic vs real) and the quality of 3D reconstructions generated from partial observations.

WHY

- **Problem:** Incomplete 3D scans, especially from mobile devices or low-cost setups, often result in fragmented geometry with missing structure
- **Goal:** To build and evaluate a basic shape completion pipeline that is small-scale and reproducible. The aim is to measure how well a simple EPN-style model performs on known synthetic geometry

RELATED WORKS

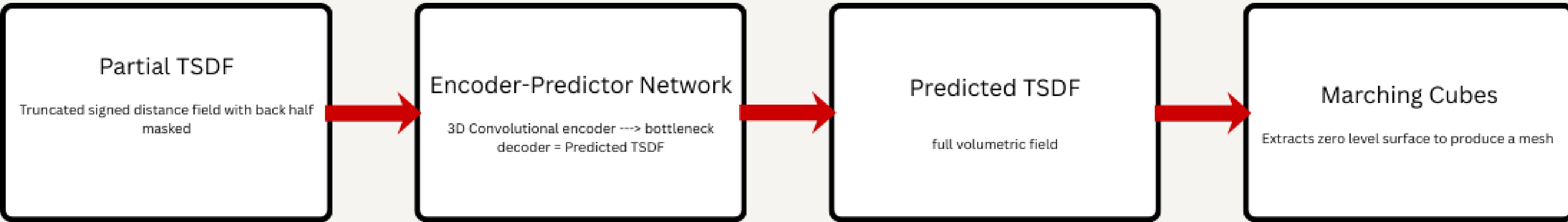
3D-EPN (Dai et al., 2017)

- Landmark work in 3D shape completion
- Showed that deep networks can infer missing geometry from partial scans
- Introduced the idea of learning a global shape prior from data
- Demonstrated strong performance on noisy, incomplete 3D inputs

How My Project Builds on This Work

- Adopts the general idea of completing shapes from partial inputs
- Uses a simplified, lightweight version suitable for small datasets
- Applies these ideas to synthetic primitives instead of large real-world scans
- Focuses on understanding model behavior at low vs. high resolution

PIPELINE



METHODOLOGY

DATA GENERATION

- A Python script was used to create simple 3D shapes (cubes, spheres, etc.)
- Each sample was converted to a TSDF volume and saved as .npz file with stored fields: **tsdf (ground truth), partial (input with missing parts), occ (occupancy grid), bounds (for alignment in 3D space)**

MESH EXTRACTION (MARCHING CUBES)

- Surface lies at $TSDF = 0$
- Algorithm finds zero-crossings and generates triangles
- Converts voxel grid to world coordinates
- Cleans the mesh to remove small floating fragments

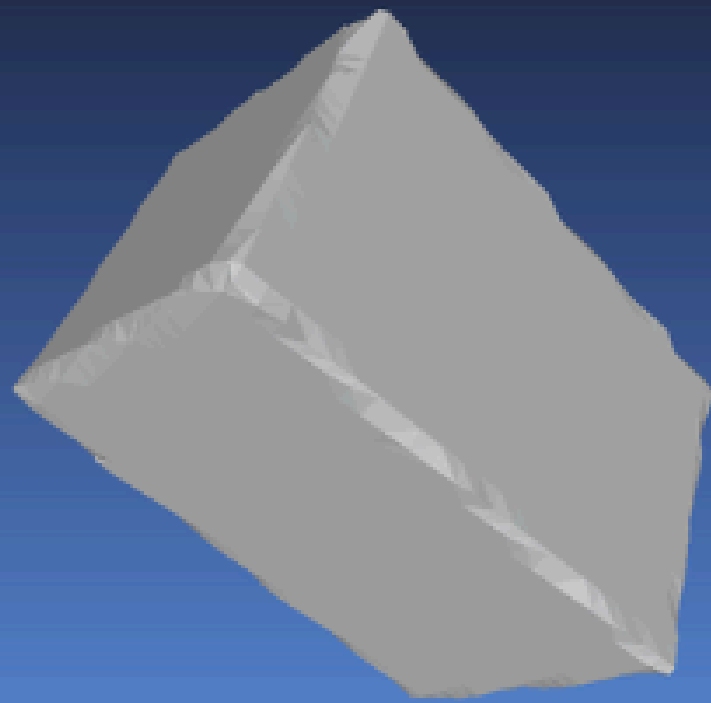
MODEL OVERVIEW

- Simple Encoder–Predictor Network (EPN)
 - Encoder: Extracts 3D features from the partial TSDF
 - Bottleneck: Forms a global “shape code”
 - Decoder: Reconstructs the full TSDF prediction

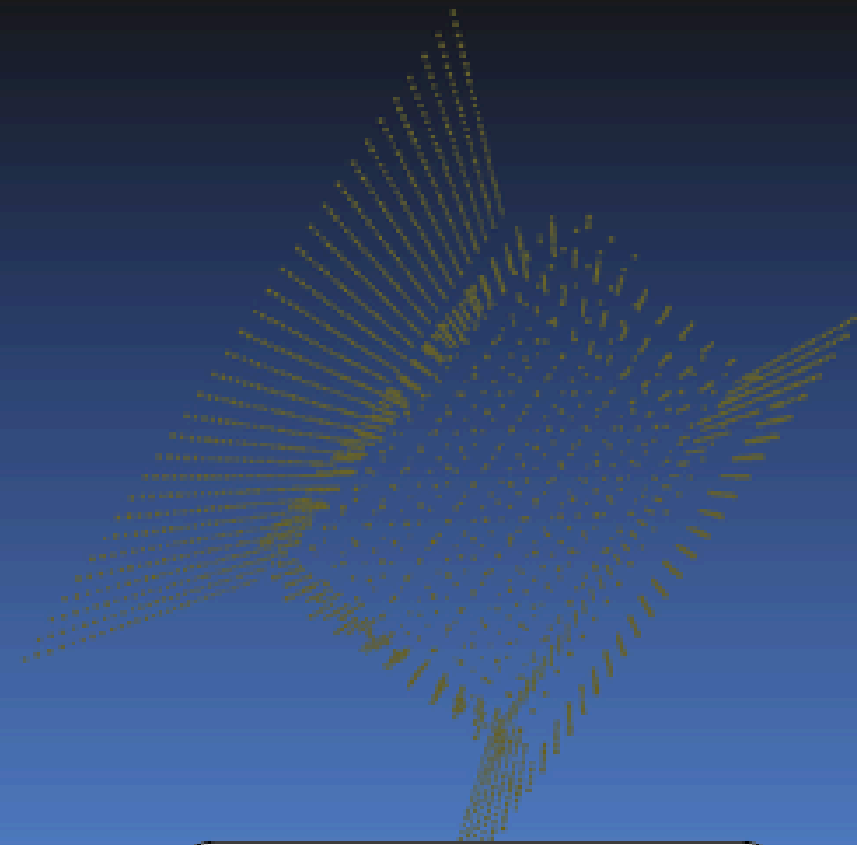
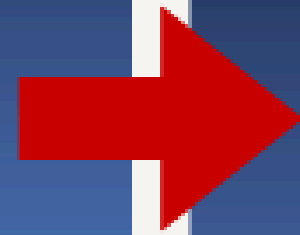
IMPROVEMENTS

- Larger data set (400 training models)
- Higher resolution (64^3) → lower loss, more detail
- Added BatchNorm3d → much more stable and consistent training
- Increased model capacity (16→32 channels) → improved complex shapes
- Training refinements → lower LR, gradient clipping, mixed precision

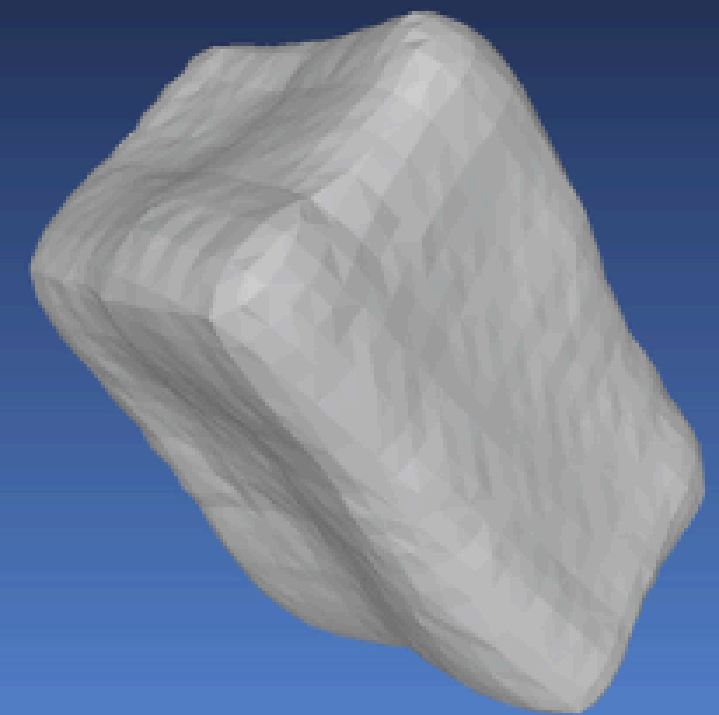
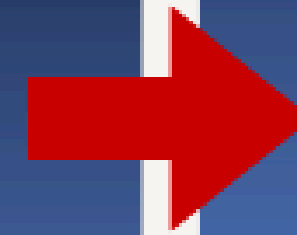
INITIAL RESULTS



Ground Truth



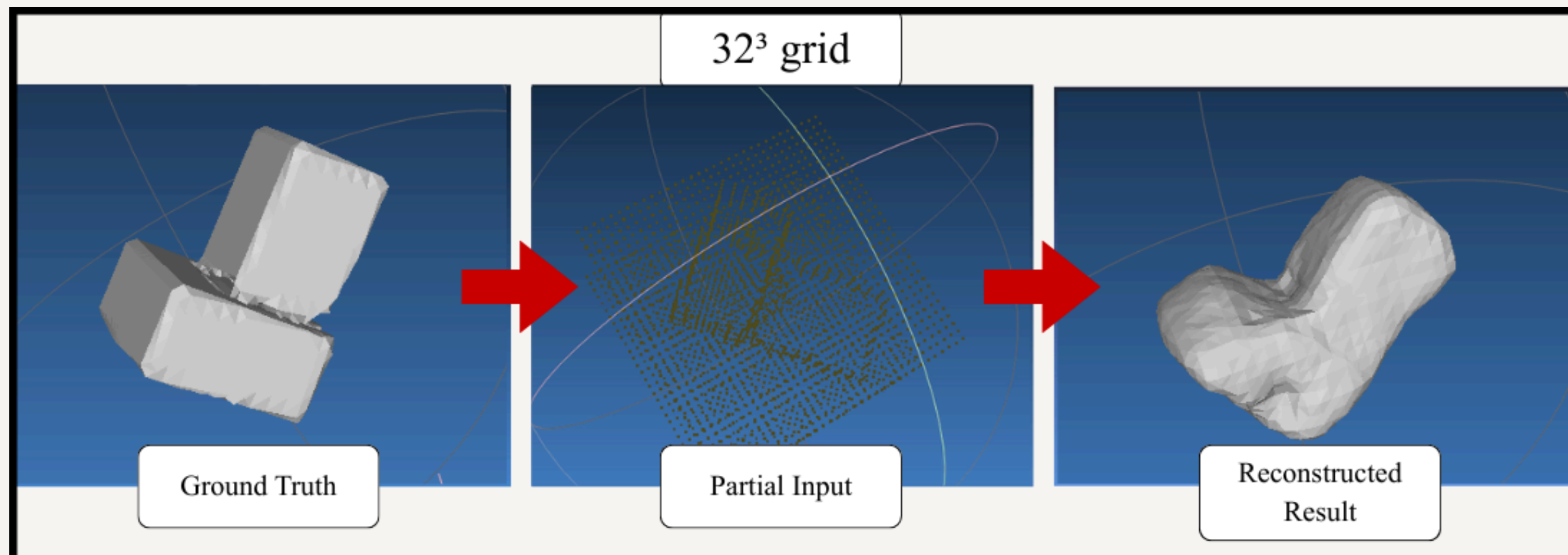
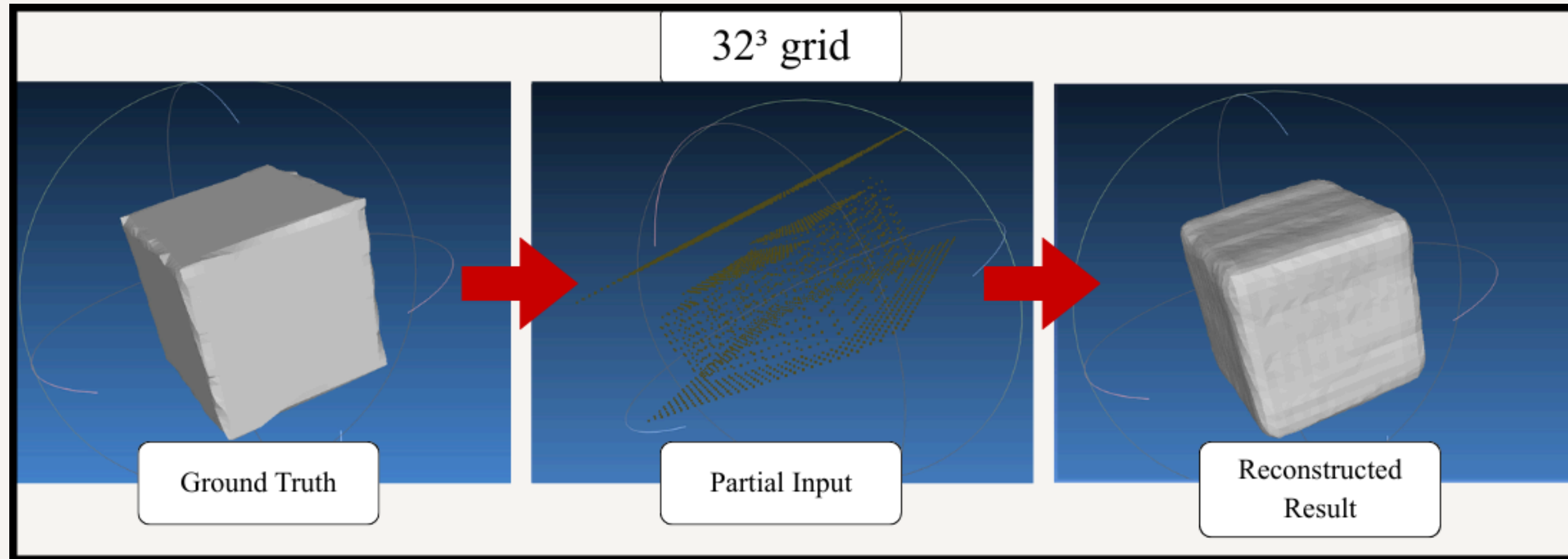
Partial Input



Reconstructed
Result

BASELINE MODEL — 32^3 RESOLUTION

RAN WITH IMPROVED PRIMITIVES



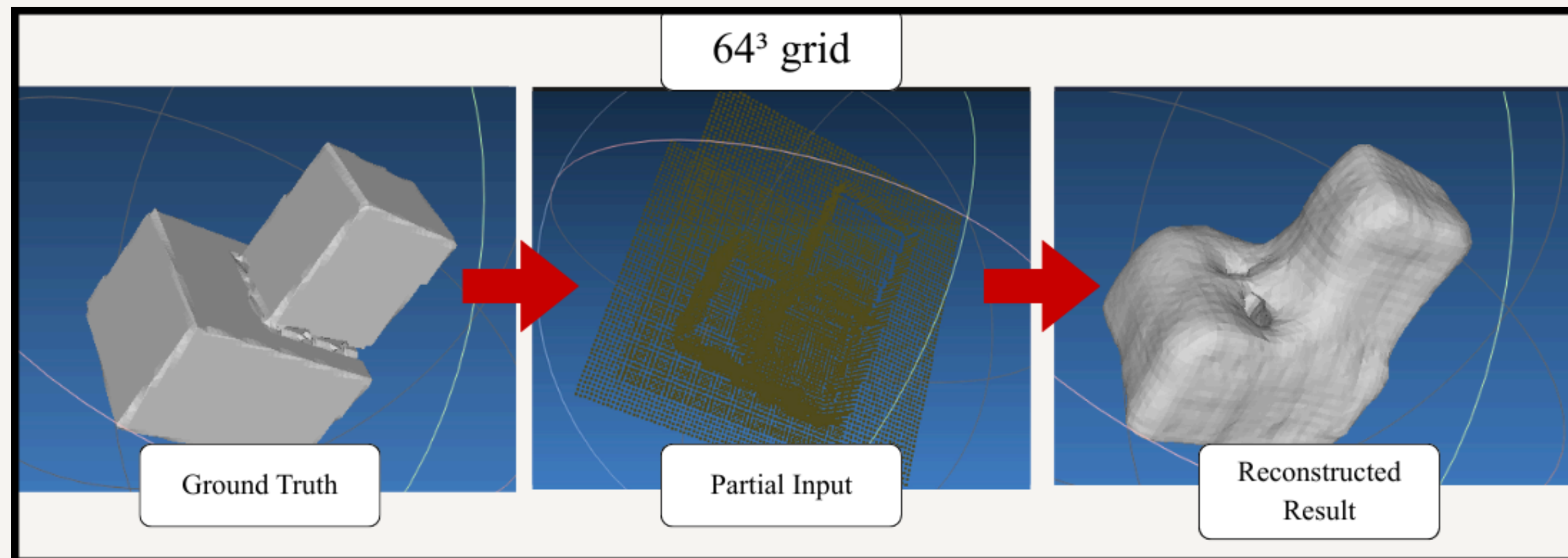
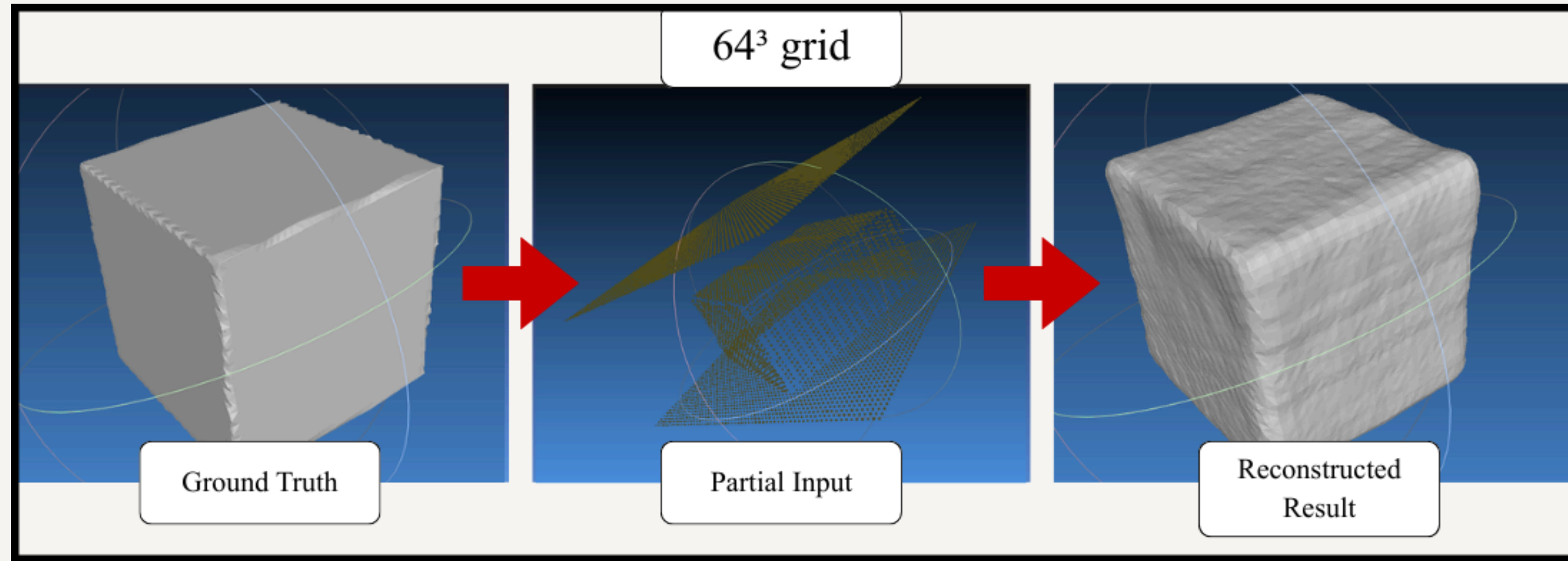
TRAIN LOSS: $0.701 \rightarrow 0.136$

VALIDATION LOSS: 0.104

IMPROVEMENT: 80.6%

TIME TO TRAIN: 13 MIN

RESOLUTION UPGRADE: $32^3 \rightarrow 64^3$



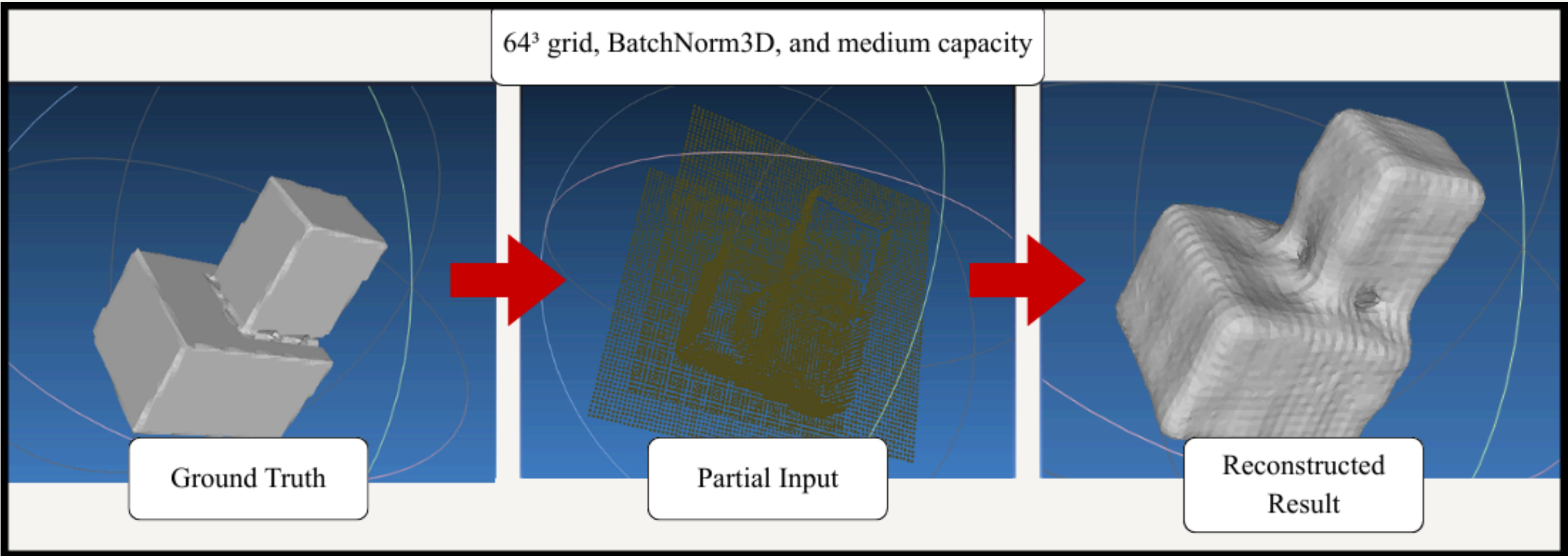
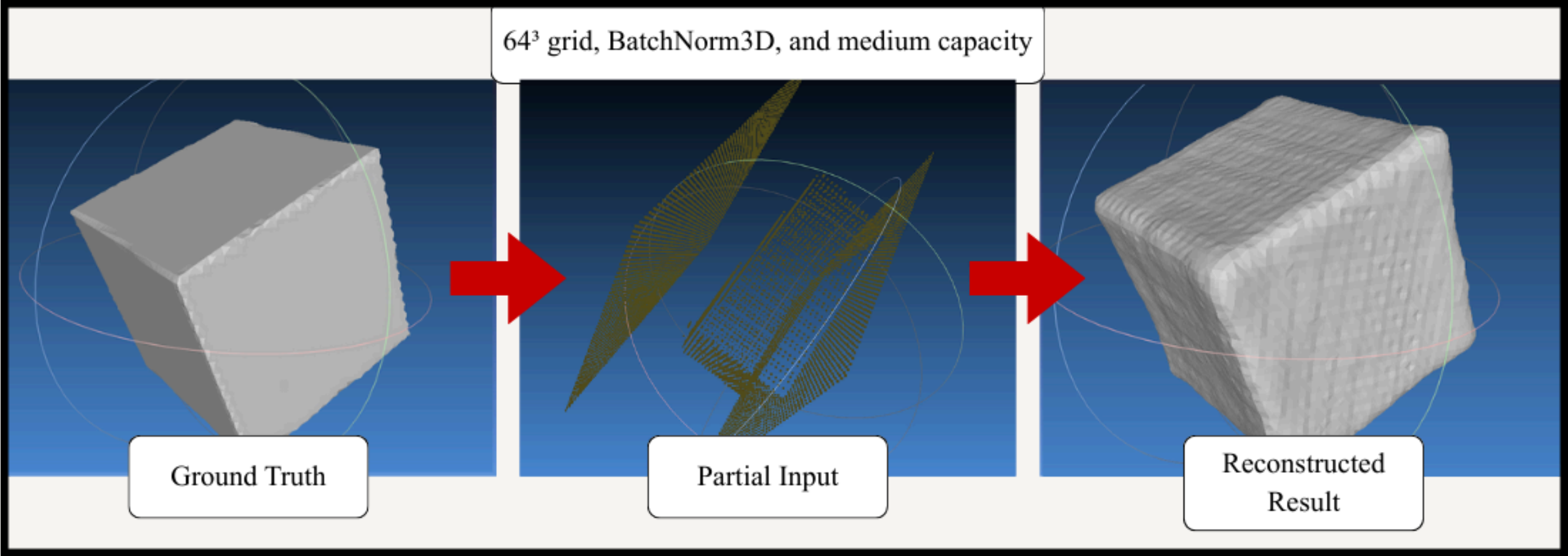
TRAIN LOSS: $0.462 \rightarrow 0.061$

VALIDATION LOSS: 0.049

IMPROVEMENT: 86.8%

TIME TO TRAIN: 2 H 14 MIN

FINAL MODEL — 64^3 + BATCHNORM3D + LARGER NETWORK



TRAIN LOSS: 0.093 → 0.001
VALIDATION LOSS: 0.007
IMPROVEMENT: 98.4%
TIME TO TRAIN: 1 H 7 MIN

PLAN FOR COMPLETION/EVALUATION

EVALUATION METRICS USED

- Training Loss (MSE)
- Validation Loss
- Reconstruction Detail
- Mesh Quality Check

COMPARISON PLAN

- Evaluate model performance across:
 - Different Resolutions
 - 32^3 (baseline)
 - 64^3 (higher detail)
 - Different Model Capacities
 - Small model (16 channels)
 - Medium model with BatchNorm (32 channels)
 - Training Stability Metrics
 - Convergence speed
 - Overfitting vs. underfitting behavior
 - Consistency across validation samples

FINDINGS

- 64^3 models significantly outperform 32^3 in both loss and reconstruction detail
- BatchNorm + increased channels improves stability, reduces validation loss, and produces 3–4× more detailed meshes
- Mesh failures did not occur after fixing truncation and checkpoint loading
- Best model achieved `val_loss = 0.007` with 98% reduction in training error

DEMONSTRATION PLAN

- Show side-by-side comparisons for each stage:
 - Ground-Truth Mesh
 - Partial TSDF
 - Reconstruction

REFERENCES

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